MIL-E-1/22H
5 March 1973
SUPERSEDING
MIL-E-1/22G
24 January 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPES 1B35A, 1B37A, 6162, AND 5864

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: ATR, frequency: 1B35A --- Fo = 9,300 MHz, incident power 250 kw
1B37A --- Fo = 8,750 MHz
6162 --- Fo = 9,080 MHz
5864 --- Fo = 9,375 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	TA	Alt
Unit:	kw	---	°C	ft
Maximum:	250	0.001	+100	10,000
Minimum:	4	---	-40	---
		(See note 1)		

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS: See note 2

Parameter:	F	tp	prr	Du	Incident power	σ'
Unit:	MHz	μs	pps	---	kw	---
Test condition 1:	Fo ± 3%	1.0 ± 0.1	1,000	0.001	200	1.2 (max)
Test condition 2:	Fo ± 3%	1.0 ± 0.1	1,000	---	20	1.03

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
1027	Temperature cycling (nonoperating)	-		---	---	---	---
4482	Tuning susceptance	-	Fo ± 0.1% MHz	b = B/Yo	---	±0.06	---
4486	Firing time	3		t	---	10	sec
4488	Arc loss	3		La	---	0.8	dB
	Quality conformance inspection, part 2						
4101	Dielectric material strain	-		---	---	---	---
4484	Normalized conductance	-	Fo ± 0.1% MHz	g = G/Yo	---	0.1	---
4471	Recovery time	-	Test condition 1	t	---	10	µs
	Quality conformance inspection, part 3						
---	Life test	-	Group D; test condition 1				
	Type 1B35A	-		---	500	---	Hrs
	Type 1B37A	-		---	500	---	Hrs
	Type 6162	-		---	500	---	Hrs
	Type 5864	-		---	1,000	---	Hrs
---	Life-test end points:						
4488	Arc loss	3		La	---	1.0	dB
4484	Normalized conductance	-	Fo ± 0.1% MHz	g = G/Yo	---	0.1	---
4482	Tuning susceptance	-	Fo ± 0.1% MHz	b = B/Yo	---	±0.06	---
4471	Recovery time	-	Test condition 1; VSWR > 10 dB	t	---	20	µs
	Periodic-check tests						
1027	Temperature cycling life-test end point	3	Group D; 10 cycles (min)	---	---	---	---
4021	Degradation due to vibration	-		---	---	---	---
4461	Loaded Q (ATR tubes)	-		QL	---	6.5	---
4474	High-level VSWR	-	Test condition 2	σ	---	1.10	---

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NOTES:

1. For 500 hours life, the duty cycle is 0.001 at pulse width of 1.0 μs. Operation at greater than 1.0 μs pulse width at 250 kw is permissible but may result in decreased life.

2. For electrical tests, the tubes shall be mounted as shown on figure 2. A suitable gasket shall be placed between the tube and the tube seat.

3. This test shall be performed at least 168 hours after pumping, and at least 24 hours after any previous discharge.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activity:
DSA - ES

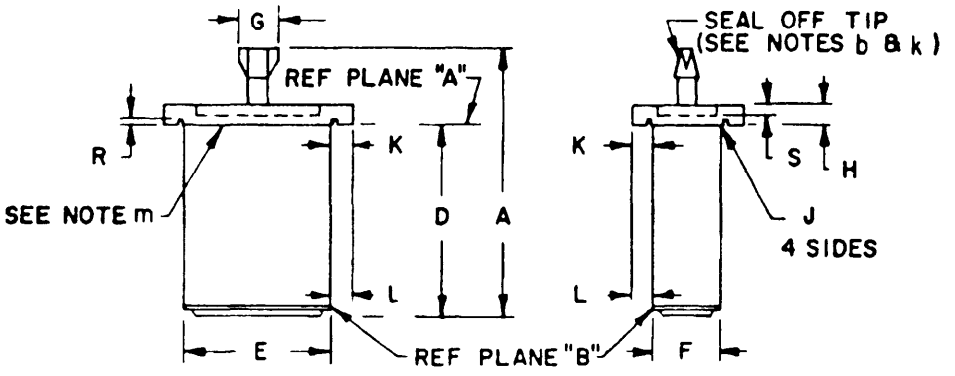
User activities:
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Navy - AS, OS, MC, CG, SH
Air Force - 11

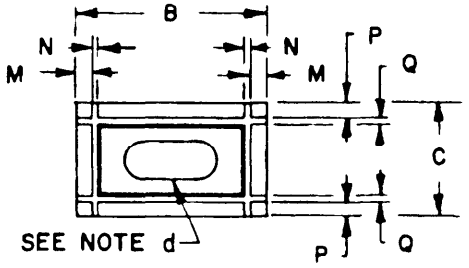
Preparing activity: Navy - EC

Agent: DSA - ES

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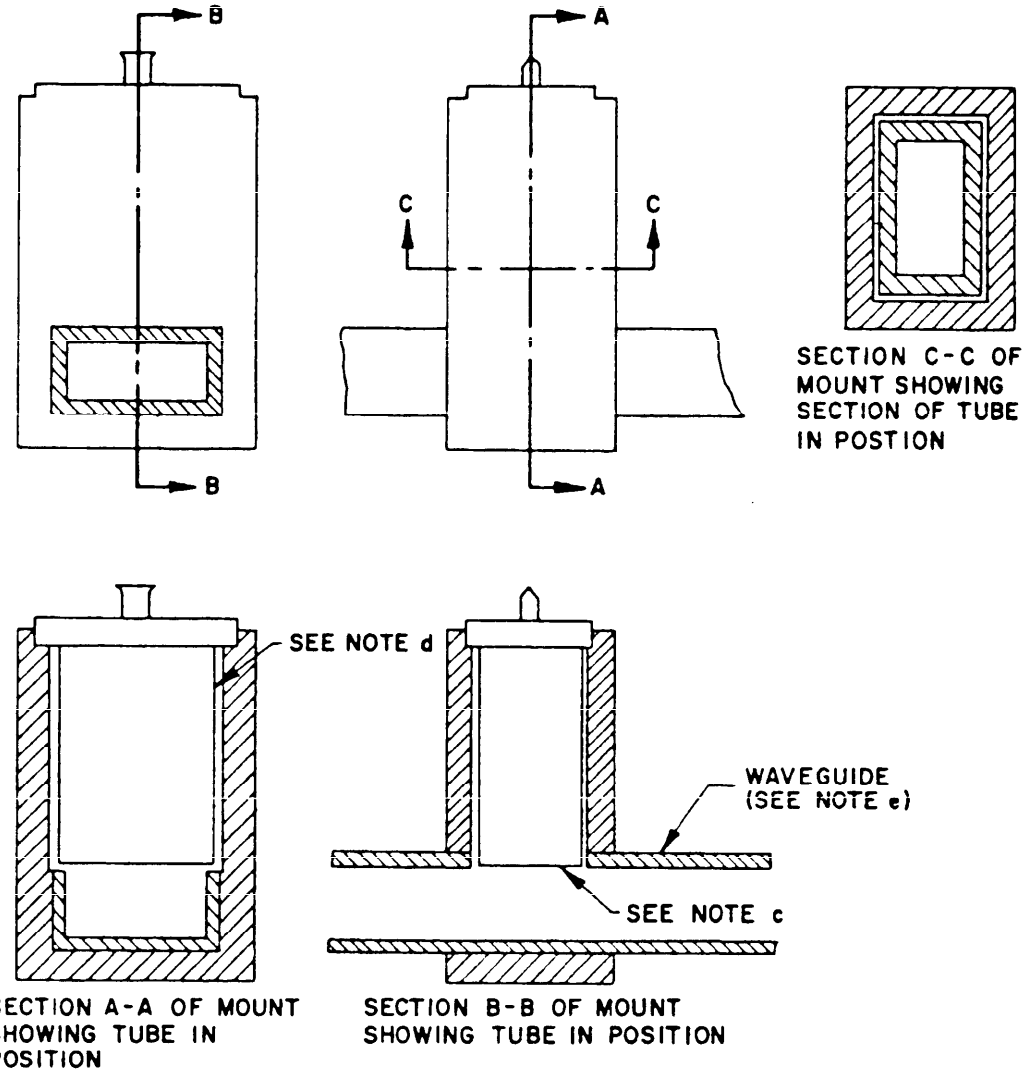
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 1		
B	1.297 (32.94)	1.303 (33.10)
C	.797 (20.24)	.803 (20.40)
D	1.296 (32.92)	1.302 (33.07)
J		.020 (.51)
K	.142 (3.61)	
L	.138 (3.53)	
S		.078 (1.98)
Dim M through R for 1B37A only		
M	.103 (2.62)	.111 (2.82)
N	.030 (.76)	.040 (1.02)
P	.103 (2.62)	.111 (2.82)
Q	.030 (.76)	.040 (1.02)
R	.035 (.89)	.045 (1.14)
Quality conformance inspection, part 2		
A		1.813 (46.05)
E	.990 (25.15)	1.010 (25.65)
F	.490 (12.45)	.510 (12.95)
G		.250 (6.35)
H	.117 (2.97)	.133 (3.38)

NOTES:

a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. Tubulation shall lie within a circle .38 (9.65 mm) diameter maximum located from centerlines of flange. Qualification inspection required.
c. Entire tube shall be plated 100 MSI minimum silver, or equivalent. Qualification inspection required.
d. Centerlines of window shall coincide with corresponding centerlines of tube body within .015 (.38 mm) measured in plane of window. Quality conformance inspection, part 2, required.
e. Two gaskets as specified on Drawing 191-JAN shall be supplied unmounted. Quality conformance inspection, part 2 required.
f. Dimension K shall be measured .12 (3.05 mm) from reference plane A.
g. Dimension L shall be measured .12 (3.05 mm) from window plane.
h. Reference plane A shall be normal to reference plane B.
j. Flat end plate and dimpled window optional.
k. Seal-off cap over exhaust tubulation optional.
m. Position of gasket when mounted.

FIGURE 1. Outline drawing of electron tube types 1B35A, 1B37A, 6162 and 5864.

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NOTES:
a. Design of holder optional.
b. Tube shall be held firmly in position by means of spring or screw clamps.
c. Bottom of tube shall be flush with inside surface of waveguide within ± .004.
d. Clearance between tube walls and inner surface of mount shall be .030 to .040.
e. Waveguide shall be RG-52/U for the 1B35A, 1B37A, and 6162; and RG-51/U for the 5864. It is recommended that the RG-52/U waveguide be pressurized to prevent pulse power breakdown at 250 kw. The RG-52/U and RG-51/U waveguides shall be in accordance with MIL-W-85/1.

FIGURE 2. Test mount.

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